Is an All-White Opening Really Disadvantageous in Two-Player Da Vinci Code?

A Computational Study of Initial Color Composition, Pooling Signals, Joker Placement, and First-Exchange Value

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Abstract—Da Vinci Code (Coda) is an imperfect-information deduction game in which players infer hidden numbers from visible color sequences and revealed tiles. This study fixes the first attacker's initial code at two white and two black tiles (2+2) and evaluates an all-white four-tile opening for the counterattacker over the first attack–counterattack exchange. Two analytical layers are used. First, a one-step maximum-a-posteriori (MAP) baseline removes jokers and strategic signaling, enabling a common-condition comparison between 2+2 versus 2+2 and 4W versus 2+2. Second, a signal-aware solid 4W policy incorporates jokers, pooling, extreme-value declarations, and joker placement, and is evaluated by exact enumeration of 180,180 microstates. Under the common baseline, the first attacker's opening loss D=L-G was 0.14772 for 2+2 versus 2+2 and 0.17232 for 4W versus 2+2. Under the separate solid-policy model, Hero's first-hit probability was 508/2145 and Villain's counter-hit probability was 1733/6435, yielding D=19/585>0. Counterintuitively, an all-white code appears vulnerable because information is concentrated in one color, yet the branch in which Villain obtains the white joker with probability 4/13 and places it at the far left blocks Hero's pooled W0 attack and reverses the aggregate sign. This report calls the phenomenon the "All-White Paradox" as a descriptive, author-defined term. The reported values are fixed-policy first-exchange benchmarks, not full-game win probabilities or a GTO solution.

Index Terms—Da Vinci Code, Coda, imperfect-information games, initial color composition, pooling, joker placement, posterior probability, exact state enumeration

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I. Introduction

1. Background

Da Vinci Code (Coda; Algo in Japan) is a deduction game in which colored number tiles are ordered from low to high and players infer hidden values in an opponent's code. Colors and positions are visible while numbers remain hidden, so a declaration is both a probabilistic choice and a potential signal about the declarer's private information. The official rules allow a player to choose any combination of light and dark tiles for the initial four-tile code, making initial color composition a strategic variable [1].

A natural player intuition is that 2+2 is the safest composition and that concentrating all four tiles in one color is informationally fragile. If Villain holds four white tiles, every white tile drawn by Hero removes one candidate directly from Villain's possible white numbers. The starting hypothesis was therefore that an all-white opening should be easy to read and disadvantageous.

2. Research Question

> **Primary research question**
>
> With Hero's initial composition fixed at 2+2, is Villain's all-white four-tile opening actually disadvantageous over the first attack–counterattack exchange?

The study does not directly solve the full game or a complete equilibrium. It isolates the smallest analyzable interval—one Hero attack followed by one Villain counterattack—and computes the probability that each side reveals one opposing tile. This truncation discards later dynamic value but permits a reproducible comparison of color composition and information order.

3. Related Work and Position of This Report

Publicly available adjacent work includes a course project that modeled Coda using an MDP and reinforcement learning [3], and a Japanese technical implementation that used average information gain to select attacks and stopping decisions [4]. These works address general action selection or whole-game information gain rather than the narrow comparison of initial 4W and 2+2 compositions over the first exchange. This report is therefore positioned as a limited computational study of opening composition and signaling, not as a complete game solver.

4. Contributions

- Separate standard rules from the research-specific variant so that readers unfamiliar with the game can follow the model.
- Replace colloquial language with the formal concepts of a one-step maximum-a-posteriori guess and a pooling policy.
- Compare 2+2 and 4W under a common minimal baseline, then evaluate a separate signal-aware solid 4W policy by exact enumeration.
- Explain the counterintuitive positive opening value of 4W through the author-defined "All-White Paradox."
- Present an exploratory meta-game in which publication of the strategy changes the opponent's best response.

II. Game Rules and Formalized Variant

1. Standard Da Vinci Code Rules

The standard set contains white and black tiles numbered 0 through 11, plus one color-matched joker (dash tile) for advanced play. In a two-player game, each player takes four tiles, hides the numbers from the opponent, and orders them from left to right. When equal numbers of different colors coexist, the black tile is placed to the left of the white tile. On a turn, a player draws one tile and declares the color and number of a specific opposing position. A correct declaration reveals the opponent's tile and may be followed by another attack; an incorrect declaration forces the player to reveal the tile drawn that turn and ends the turn [1].

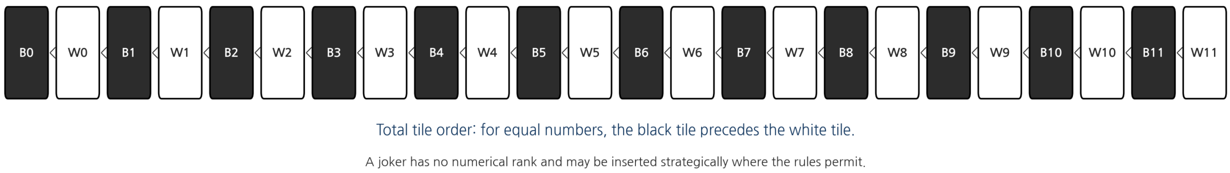

Fig. 1. Total ordering induced jointly by number and color.

2. Core Intuition for a New Reader

An opponent's tile backs reveal their colors, so a player simultaneously uses color, left–right order, held tiles, and previously revealed tiles. A tile held by the player cannot be in the opponent's code. An attack is therefore not a blind random guess: it is a choice among position–color–number triples after enumerating all opponent codes consistent with the current information.

3. Research-Specific Rules

TABLE I. Research-specific opening rules extracted from the standard game

Item	Fixed rule	Interpretation
Players	Two	Hero attacks first; Villain counterattacks.
Initial code	Hero fixed at 2+2	Villain uses either 2+2 or four white tiles.
Numbers	Random	Only color composition is selected.
Attacks	One each	No continued attacks after a hit.
Evaluation interval	One opening exchange	Hero draw/attack followed by Villain draw/counterattack.
Public information	Color order, revealed tiles, draw position	Use of declaration signals depends on the model.
Jokers	Excluded in Model A; included in Model B	In Model B, a revealed joker is conservatively assigned zero extra numerical information loss.
Win condition	Not evaluated	The metric is expected revealed-tile difference, not full-game win probability.

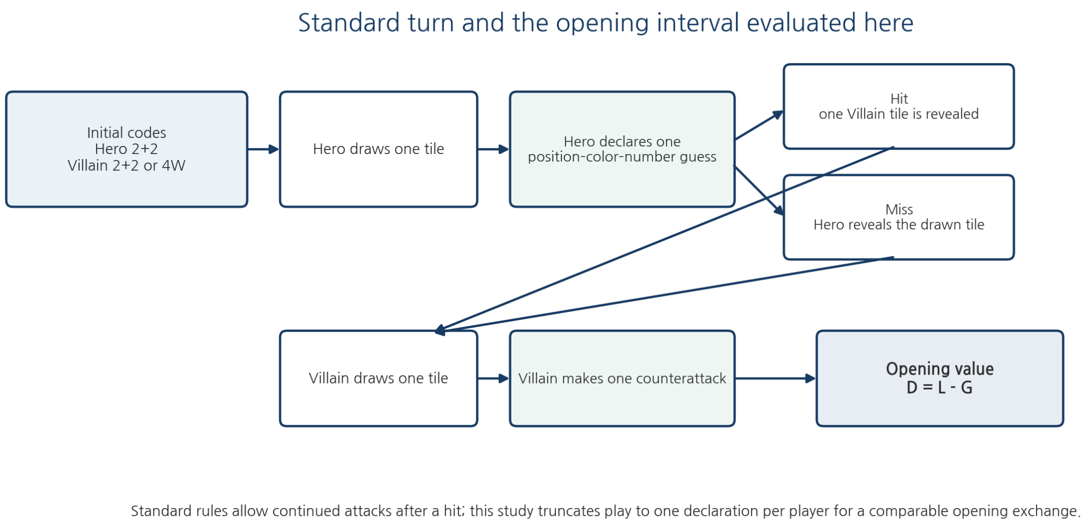

Fig. 2. First attack–counterattack interval evaluated in this report.

4. Roles and Notation

TABLE II. Notation and key concepts

Symbol/term	Definition	Meaning here

Symbol/term	Definition	Meaning here
Hero (H)	First attacker	The names indicate order, not moral roles.
Villain (V)	First counterattacker	Observes Hero’s outcome before responding.
Wk / Bk	White k / Black k	For example, W0 is white 0 and B11 is black 11.
G	P(Hero first hit)	Probability that Hero’s first declaration is correct.
L	P(Villain counter-hit)	Probability that Villain’s first counterattack is correct.
D	L-G	Hero opening loss; D>0 favors the counterattacker.
MAP guess	Maximum-a-posteriori guess	An immediate declaration with maximal posterior hit probability.
Pooling	Same action across private states	Reduces information leakage from the action itself.

Terminology

The colloquial Korean expression used during brainstorming is not retained in the formal report. It is standardized as a “one-step maximum-a-posteriori (MAP) guess,” which is not a random guess.

III. Logic Behind the Research Hypotheses

1. Equal Expected Joker Count Does Not Imply Equal Distribution

The expected number of jokers in four initial tiles is 4/13 across the considered color compositions. With four white tiles, the white joker is included with probability 4/13; with 2+2, the inclusion probabilities of the two color-specific jokers sum to the same expectation. However, the color in which joker probability is concentrated and the probability of holding at least one joker differ. Expected joker count alone is therefore insufficient for evaluating an opening composition.

$$\mathrm{E}[\text{number of jokers}] = 4/13 \ \text{ for } \ 4\mathrm{W} \ \text{ and } \ 2{+}2$$

2. State-Dependent Actions Become Tells

If Hero declares W0 only when W0 is absent and switches to W11 when W0 is held, Villain can infer W0 ownership from the declaration alone. The same issue applies to draw color: drawing white only with a white joker and black otherwise makes the black draw a public signal that Hero lacks the white joker. A stable one-shot policy should therefore use the same action across as many private hand types as possible.

Pooling principle

Hero’s action should remain as constant as possible. If the action changes with the private hand, Villain receives information from the choice itself before observing the outcome.

3. Why Hero Draws White

- Because Villain holds four white tiles, a white draw directly removes one number from Villain’s candidate set.
- A black draw does not shrink Villain’s white candidate set and may signal that Hero lacks the white joker.
- After a failed black draw, the revealed black tile gives Villain a new reference point for attacking Hero’s black code.
- If Hero draws and later reveals the white joker, the reveal carries almost no additional numerical range information; the conservative extra-loss assignment is therefore zero.

4. Why W0 Is the Default Declaration

The simplest attack on four white tiles is to declare the smallest remaining white candidate at Villain’s leftmost position. If n candidates remain and Villain holds four of them, the probability that the minimum candidate is included is 4/n; when included, ordering forces it to the left edge. However, changing away from W0 when Hero holds W0 creates a tell, so W0 pools ordinary attacks with defensive bluffs.

5. W1 and W11 Are Not Informationally Symmetric

On the number line alone, 1 and 11 appear roughly symmetric, but the actual order is B0<W0<B1<W1<…<B11<W11. Revealing W1 strongly restricts which black tiles can lie to its left—essentially B0 and B1—whereas W11 does not compress low-end black information in the same way. In the rare state where Villain holds both W1 and W11, moving to W2 would signal both holdings. A defensive W1 pooling declaration is therefore the more conservative choice.

6. The All-White Paradox

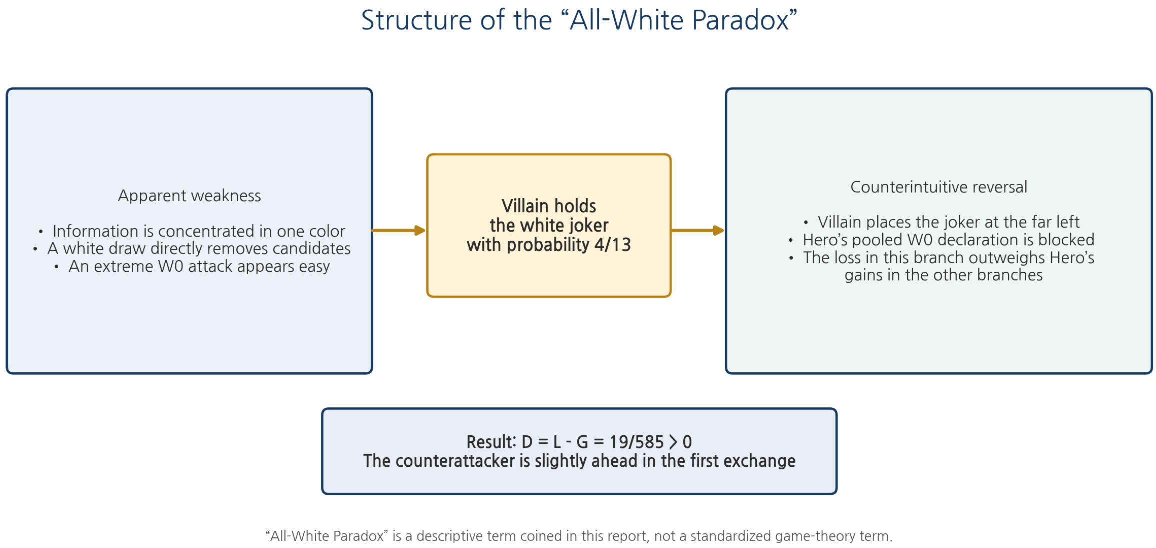

Fig. 3. Logical structure of the author-defined "All-White Paradox."

Four white tiles are vulnerable because white candidates are removed quickly. Yet in the 4/13 branch where Villain holds the white joker, Villain can place it at the far left against Hero's pooled W0 policy. This is a best response to a fixed Hero policy, not a joint GTO solution. The negative contribution of this branch exceeds Hero's gains in the remaining branches and reverses the aggregate sign in favor of the counterattacker.

IV. Analytical Design and Value Function

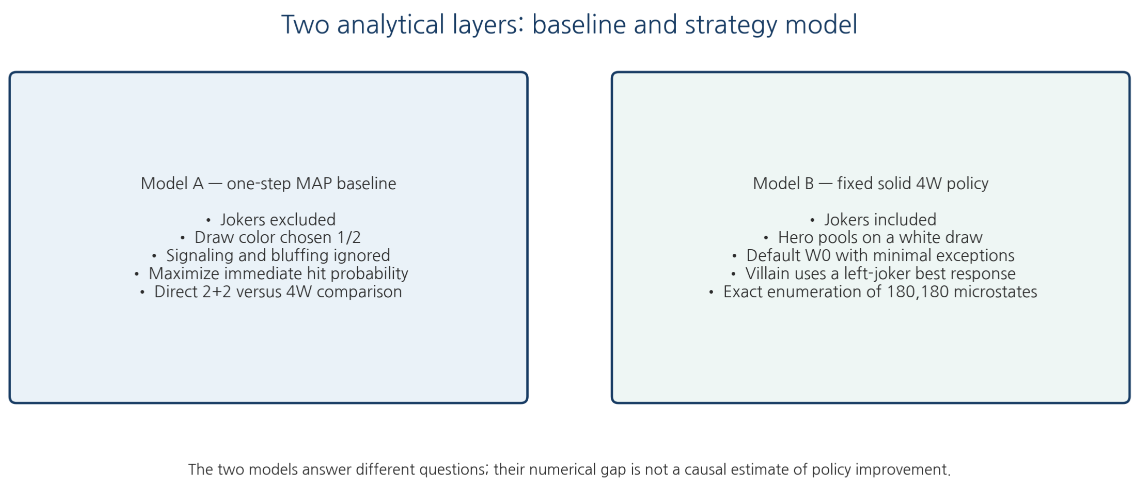

Fig. 4. The two analytical layers used in this report.

1. Common Value Function

Let G denote Hero's first-hit probability and L Villain's first counter-hit probability. Hero's opening loss D is the expected number of Hero tiles exposed by Villain minus the expected number of Villain tiles exposed by Hero.

$$D = L - G \;\; ; \;\; U\_H = G - L = -D$$

D>0 means that Villain leads in expected revealed tiles over the first exchange.

A generic Hero miss is not assigned an immediate value of -1. Hero begins with four hidden tiles, temporarily reaches five after drawing, and reveals only the new tile after a miss; the original four remain hidden. Loss is therefore measured through Villain's actual counter-hit probability L rather than by treating every miss as a full tile loss.

2. Model A: One-Step MAP Baseline

- Use only the 24 numbered tiles and exclude jokers.
- Hero and Villain choose white or black draw with probability 1/2 each.
- Each player enumerates opponent codes consistent with current information and selects the position–color–number declaration with maximum immediate hit probability.
- Ignore signaling from actions, bluffing, long-term information value, and continued attacks.
- Compute Hero's G exactly and estimate Villain's L with 2,000,000 Monte Carlo trials per condition.

3. Model B: Fixed Solid All-White Policy

- Include the white joker among the 13 white tiles.
- Hero draws white from an initial 2+2 code and uses a pooled W0 default policy with minimal boundary exceptions.
- When Villain holds the joker, it is placed at the far left to block W0; after Hero's outcome, Villain draws black and counterattacks Hero's white code.
- Exactly enumerate 180,180 microstates defined by Villain's four white tiles, Hero's initial two white tiles, and Hero's newly drawn white tile.

4. What Can and Cannot Be Compared Directly

Methodological caution

The two Model A conditions are directly comparable because they share the same rules. Model B includes jokers and signaling, so subtracting 19/585 from 0.17232 and calling the gap a percentage "policy improvement" would be invalid.

V. Derivation of the Solid All-White Policy

1. Hero's Draw Policy

Hero draws white regardless of whether the white joker was initially held. This choice directly removes a Villain candidate and prevents a black draw from announcing that Hero lacks the white joker. Possession of the black joker does not change this draw policy.

2. Hero's Declaration Policy

Hero declares W0 in ordinary states. When Hero initially holds the white joker and an ordinary white number h, then draws an ordinary white number d, a minimal set of boundary exceptions is used to avoid severe chain information.

TABLE III. Fixed Hero declaration policy

Hero information state	Declaration	Rationale
No initial white joker	W0	Default pooling across joker location and W0 ownership.
h=0, d=1	W2	Mitigates joint low-boundary disclosure of W0 and W1.

Hero information state	Declaration	Rationale
h=11, d=10	W9	Symmetric high-boundary exception.
d=0, h≠11	W11	Attack the opposite extreme when the newly drawn W0 may be exposed.
h=11, d=0	W10	Move one step inward because W11 is already held.
d=11, h≠0	W0	Maintain the opposite-extreme default.
h=0, d=11	W1	Move one step inward because W0 is already held.
All other initial-joker states	W0	Default pooling that minimizes action signals.

3. Villain’s Joker Placement

When Villain holds the white joker, it is placed at the far left as a best response to Hero’s pooled W0 declaration. Even if Villain also holds W0, the joker lies farther left and Hero’s W0 declaration misses. This is a fixed-policy best response, not a joint GTO solution.

4. Villain’s Counterattack Policy

1. Draw black after either a Hero hit or miss, avoiding further concentration of Villain’s white code.
2. Attack Hero’s white code; declare W0 whenever W0 can remain in Hero’s hidden code.
3. If W0 is already public or held by Villain, choose the more likely of W1 and W11, breaking exact ties in favor of W1.
4. If Villain holds both W1 and W11, do not move to W2; use a defensive W1 pooling bluff so the action does not reveal both holdings.
5. After a special non-W0 Hero declaration, use a posterior-maximizing counterattack favorable to Villain.

> **Nature of the policy**
> This is not a complete GTO solution. It is a deterministic solid policy designed to minimize speculative opening play and reduce action-based signaling. The report gives the exact value of this fixed policy profile.

VI. Exact State Space and Solid 4W Results

1. State Space

Villain’s four white tiles, Hero’s initial two white tiles, and Hero’s newly drawn white tile are allocated sequentially from the 13 white tiles. Each resulting microstate has equal probability.

$$\mathrm{N} = \mathrm{C}(13,4) \times \mathrm{C}(9,2) \times 7 = 180{,}180$$

Four branches for the location of the white joker

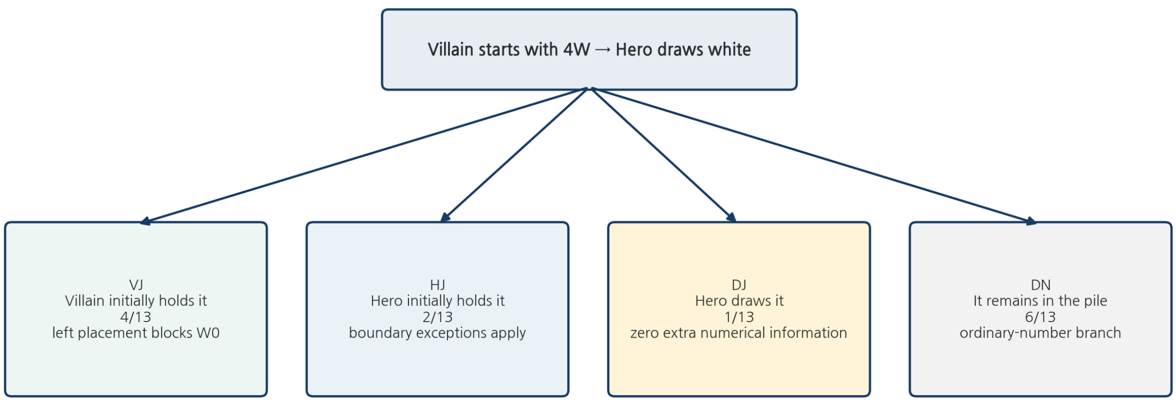

Fig. 5. State decomposition by location of the white joker.

2. Joker-Location Branches

TABLE IV. Four branches induced by the white joker

Branch	Probability	States	Meaning
VJ	4/13	55,440	Villain initially holds the white joker; left placement blocks W0.
HJ	2/13	27,720	Hero initially holds the white joker; boundary exceptions apply.
DJ	1/13	13,860	Hero obtains the joker on the first white draw.
DN	6/13	83,160	Hero draws a number and the joker remains in the pile.

3. Branch-Level Values

TABLE V. Hero hit and Villain counter-hit values by branch

Branch	p_H	q_S	q_F	L_s	Interpretation
VJ	0	—	109/440	109/440	The left joker completely blocks Hero's W0.
HJ	41/110	119/369	31/207	106/495	Includes Hero boundary exceptions and signal cost.
DJ	1/3	13/55	1/4	27/110	A revealed new joker has zero extra numerical information loss.
DN	1/3	39/110	31/110	101/330	Ordinary-number branch with the joker left in the pile.

$L_s = p_H q_S + (1-p_H) q_F$. q_S and q_F are Villain's counter-hit probabilities after a Hero hit and miss, respectively.

4. Aggregate Result

$$G_H = (4/13)\cdot 0 + (2/13)\cdot(41/110) + (1/13)\cdot(1/3) + (6/13)\cdot(1/3) = 508/2145$$

$$L_{4W} = (4/13)\cdot(109/440) + (2/13)\cdot(106/495) + (1/13)\cdot(27/110) + (6/13)\cdot(101/330) = 1733/6435$$

$$D_{solid,4W} = L_{4W} - G_H = 19/585 \approx 0.032479 > 0$$

Under the specified solid policy, the first attacker loses an expected 19/585 of a tile over the first exchange. This is not the probability of losing the game; it is the expected difference in opposing tiles revealed after one attack per player.

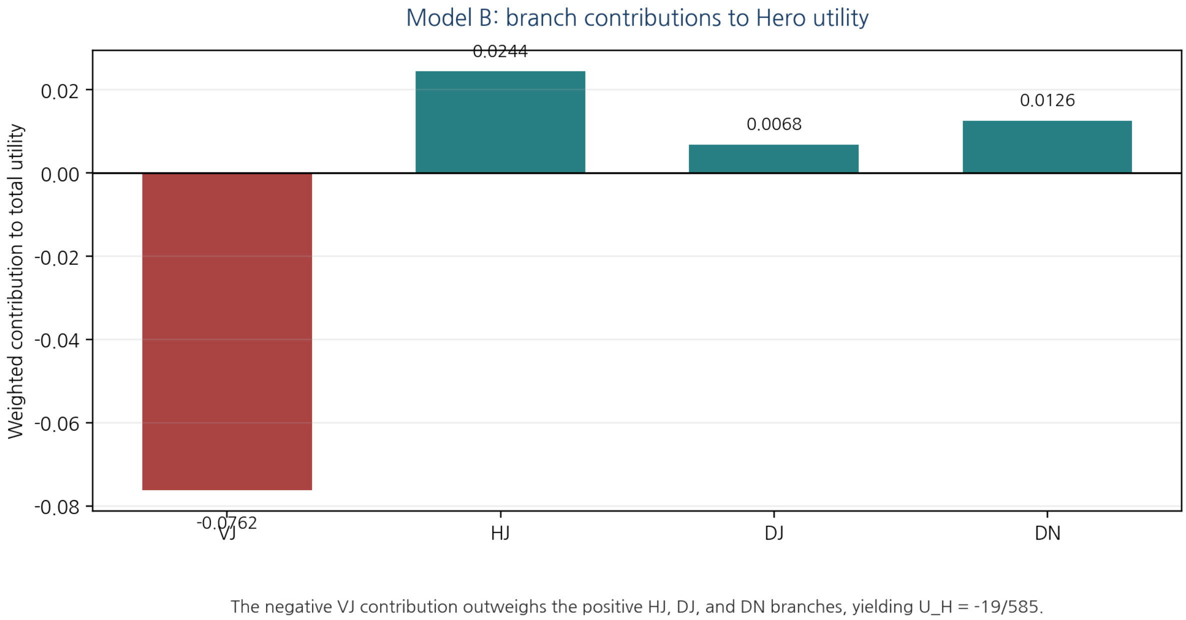

Fig. 6. Weighted branch contributions to Hero utility.

5. Why the Aggregate Sign Is Positive

Hero has positive branch utility $p_H - L_s$ in HJ, DJ, and DN. However, the weighted loss -109/1430 in the VJ branch exceeds the positive contributions of the other three branches. The small advantage of the all-white opening therefore comes primarily from concentrating the relevant joker in the same color and using it to block the pooled W0 policy—not from all-white codes being uniformly robust.

VII. Common Baseline: 2+2 versus All-White

1. Model A Results

TABLE VI. Common baseline with jokers and signaling removed

Villain code	Hero G	Villain L	Hero loss D	Verification
2+2 vs 2+2	3170971/7840800 = 0.404419	0.552139	0.147720	Exact Hero G / 2M-trial L
4W vs 2+2	19/45 = 0.422222	0.594540	0.172317	Exact Hero G / 2M-trial L
Difference: 4W-2+2	+0.017803	+0.042401	+0.024598	Common code conditions

Monte Carlo 95% CIs: D=0.147030–0.148409 for 2+2 and 0.171637–0.172998 for 4W; the difference is 0.023629–0.025566.

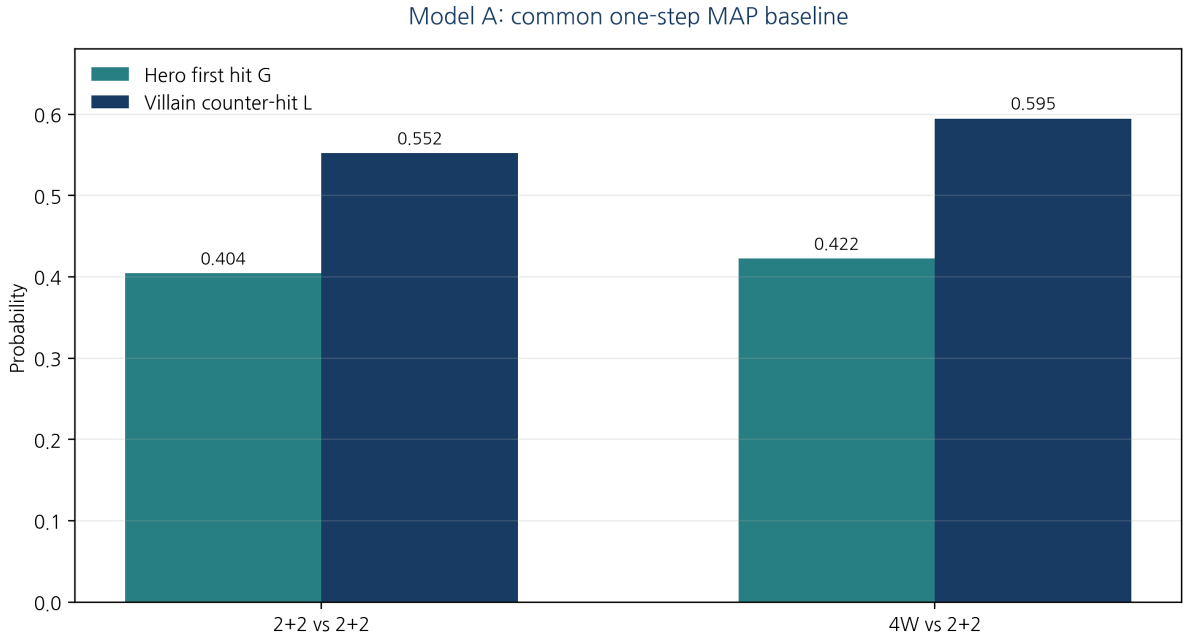

Fig. 7. Hero first-hit and Villain counter-hit probabilities under the common baseline.

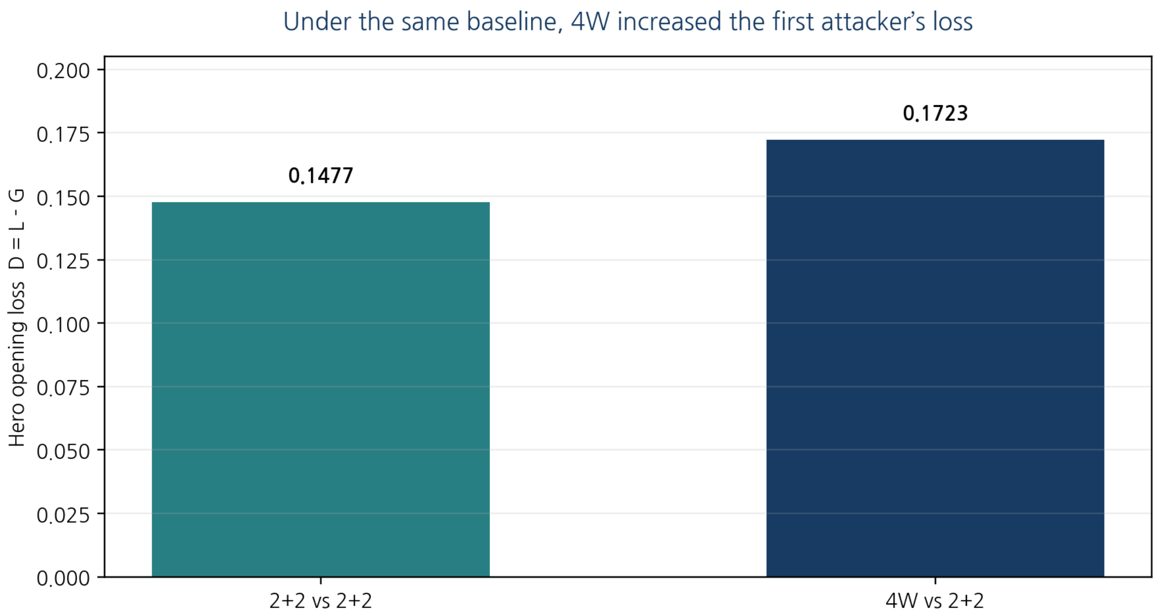

Fig. 8. Comparison of first-exchange loss D=L-G.

2. Interpretation

Even when both players use 2+2, Hero’s opening loss is positive at approximately 0.14772. The compositions are symmetric, but the information times are not: Villain observes Hero’s outcome and, after a

Hero miss, the newly revealed tile, then completes a draw before counterattacking. The value therefore serves as a baseline for the informational order effect rather than for color-composition asymmetry.

When Villain uses four white tiles, Hero’s first-hit probability also rises, from 0.40442 to 0.42222, confirming that an all-white code is easier to read. Yet Villain’s counter-hit probability rises more, from 0.55214 to 0.59454, increasing Hero’s net loss by approximately 0.02460. The all-white code gives Hero an easier first shot but produces an even stronger counterattack environment.

One-line interpretation for a new player

The right to attack first was not automatically a reward. Even under a simple immediate-probability baseline, the first counterattacker exposed more expected tiles than the first attacker.

3. Relationship Between the Baseline and Solid Policy

TABLE VII. How the three reported values should be read

Model	Assumptions	D	Correct reading
2+2 baseline	No jokers, no signals	0.147720	Baseline order effect.
4W baseline	No jokers, no signals	0.172317	4W adds loss under common conditions.
4W solid policy	Jokers and pooling included	19/585 = 0.032479	A separate exact model also yields D>0.

VIII. The All-White Paradox and the Disclosure Effect

1. Exact Meaning of the All-White Paradox

The result does not mean that concentrating one color is always safe. Hero’s first-hit probability is in fact higher against four white tiles. The paradox is that the informational weakness remains, yet the 4/13 branch in which Villain holds the white joker completely blocks the pooled W0 attack and reverses the aggregate first-exchange value. The weakness is not removed; it is outweighed by a nonlinear defensive branch.

2. Publication Can Undermine the Strategy

When Hero does not know Villain’s left-joker policy, W0 is a stable declaration. Once the report becomes common knowledge, however, Hero can ask whether the opponent is following the published placement. In a simple information state, let r be Hero’s belief that Villain places the joker on the far left when holding it. The immediate hit probability of a left-joker declaration is 2r/5, while that of W0 is 2/5-2r/15. The joker declaration becomes superior when r>3/4.

$$\text{Guess left joker is preferred to W0} \ \text{ iff } \ r > 3/4$$

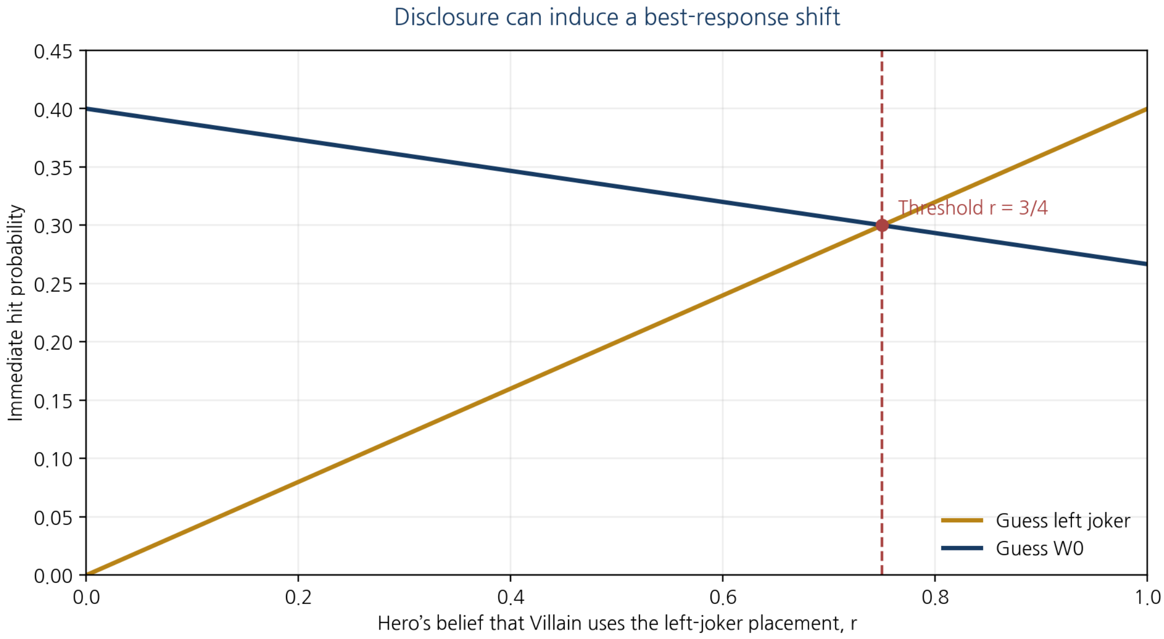

Fig. 9. Threshold at which publication induces a different immediate best response.

This report calls the phenomenon a self-undermining effect of strategy disclosure, or a disclosure-induced best-response shift. The phrase describes how a best response to a fixed opponent can lose its advantage after publication changes the opponent's policy; it is not presented as a standardized named paradox.

3. Practical Interpretation

- Against a player who uses a default extreme-value declaration, an extreme joker placement can be a strong exploit.
- If the opponent is believed to know the exploit, the same placement becomes vulnerable to a joker declaration.
- Even in a one-shot game, beliefs about what the opponent knows and follows can matter alongside card probabilities.
- The meta-game provides an educational example of how publishing a strategy can change play even without solving a full equilibrium.

IX. Limitations and Scope of Claims

TABLE VIII. Restrictions that must accompany interpretation

Limitation	Implication
No full game	Only the first attack and counterattack are evaluated; final win rate, game length, and continuation value are excluded.
No GTO proof	Mutual best responses and exploitability over the full information-set tree are not computed.
Models A and B differ	Joker and signaling assumptions differ, so their magnitudes are not a causal policy-improvement comparison.
Villain-favorable targeting	Some positional ambiguity is resolved in Villain's favor, making L a conservative upper estimate.
Simplified joker placement	Only Villain's left placement and Hero's avoidance of obvious extremes are modeled.
No human limitations	Errors, memory limits, psychology, time pressure, and opponent types are omitted.
Baseline L is simulated	Model A counter-hit values are Monte Carlo estimates and should be independently rerun with released code.

What the report supports

Under the common joker-free baseline, 4W increases Hero's first-exchange loss relative to 2+2. In the separate joker-inclusive exact solid-policy model, D=19/585>0. Thus, the intuition that the first attacker must be advantaged is not supported over this opening interval.

What the report does not support

The report does not establish that 4W is optimal for the full game, that the policy is GTO, or that strategy optimization improves loss by a precisely identified percentage.

X. Conclusion

This study tested the intuition that four white tiles must be disadvantageous because information is concentrated in one color. Under the common joker-free baseline, 4W increased Hero’s first-hit probability but increased Villain’s counter-hit probability even more, raising Hero’s net loss by approximately 0.02460 relative to 2+2. Under the separate signal-aware exact model, the solid 4W policy yielded D=19/585>0.

The main interpretation is not that all-white codes are simply strong. A genuine information-concentration weakness coexists with a strong defensive branch in which Villain obtains the white joker with probability 4/13 and blocks W0. Their interaction creates the All-White Paradox. Once the policy is published, a joker counterguess becomes possible, making the report itself part of the meta-game.

Informal conclusion

The right to attack first may not be a gift. In this first-exchange model, the player who responds after seeing one more layer of information is the one who smiles.

Future Work

1. Apply a common joker-inclusive baseline to both 2+2 and 4W for a fully apples-to-apples comparison.
2. Replace the baseline Monte Carlo counter-hit estimate with exact enumeration or dynamic programming.
3. Extend the model to continued attacks, full-game win probability, and expected game length.
4. Analyze the four visible color orders induced by a 3+1 opening as separate information sets.
5. Iterate best responses and compute exploitability toward an approximate equilibrium model.

Appendix A. One-Turn Example for a New Reader

Suppose Hero holds B2, W2, B7, and W9. The order rule gives B2<W2<B7<W9. If Hero draws W5, it is inserted to form B2<W2<W5<B7<W9. If Villain’s visible color sequence is W-W-W-W, Hero removes W2, W5, and W9 from Villain’s possible values and enumerates every four-number white code consistent with the visible order. If W0 is the smallest remaining candidate and the most likely extreme declaration, Hero points to Villain’s leftmost tile and declares W0.

On a hit, the targeted Villain tile is revealed. In this study Hero does not continue attacking; Villain immediately receives the counterattack phase. On a miss, Hero reveals the newly drawn W5. Villain then uses that public number and position, Villain’s own tiles, and Hero’s visible color sequence before responding. This explains why equal color compositions do not imply equal information at the two attack times.

Key point

The model is not “lucky guessing.” It enumerates all codes consistent with current information and selects the declaration with the largest posterior probability.

Appendix B. Computational Procedure and Reproducibility

B.1 Model A Pseudocode

```
for each trial:
    deal Hero initial 2W+2B and Villain composition
    Hero draws W or B with probability 1/2
    enumerate Villain codes consistent with Hero information
    choose a position–color–number declaration with maximal posterior hit probability
    reveal Villain tile on hit; reveal Hero drawn tile on miss
    Villain draws W or B with probability 1/2
    enumerate Hero codes consistent with all public and private information
    choose Villain MAP counterattack
estimate G, L, and D=L-G
```

B.2 Model B Exact Enumeration

Model B iterates over all C(13,4)×C(9,2)×7 white-tile allocations and applies the fixed Hero policy, hit/miss outcome, and Villain counterattack in every microstate. Cross-checks include 55,440+27,720+13,860+83,160=180,180 total states. Hero succeeds in 42,672 states and Villain counterattacks successfully in 48,524 states, reproducing 508/2145 and 1733/6435.

B.3 Recommendations Before External Submission

- Release the Monte Carlo baseline script with random seeds and software environment.
- Independently reimplement the 180,180-state enumeration.
- Separate standard rules from research-specific truncation as explicit code constants.
- Use exact fractions as primary results and decimals only as explanatory supplements.

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